

[LEDGER](#) · 2026-08-27 · INDUSTRIAL BIOTECHNOLOGY / FLAVOR & FRAGRANCE

Natural Gamma-Decalactone and Fruity Aromas via Solid-State Fermentation of **Ceratocystis fimbriata**

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Yarrowia lipolytica submerged liquid fermentation of castor oil
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Entry application	Natural Fruity/Peach Flavor Extract for clean-label beverage formulations and premium vegan cosmetics
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Measured advantage	none
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Time to first revenue	24 months
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Gross margin at parity	-283.8%
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Startup CapEx	\$350,000
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Capital productivity	0.03x
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Regulatory risk	Moderate — qualification costs reported (\$250,000 estimated)
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Why it failed

- Voided by: Capital and time to revenue, Economics undisclosed, Negative control / sub-2x, Entry application mismatch, Form-factor / handling regression
- Margin fails the operating-cost check

Assessment

The concept fails key commercial and technical criteria due to negative gross margins (-341%), high CapEx (\$350,000), long time to first revenue (24 months), and severe product/performance mismatch. The report explicitly demonstrates and concedes complete operational failure against the commercial *Yarrowia lipolytica* incumbent.

Also flagged

- Evidence rests on non-peer-reviewed sources
- Price parity asserted, not sourced

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-08-27 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

The Absence Audit

Method

Ledger

Free studies

Track record

Terms

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